

# **COSC 594**

## **Policy Optimization (II)**

**Dongsheng Ding**

**Department of Electrical Engineering and Computer Science**

**University of Tennessee, Knoxville**

# Outline

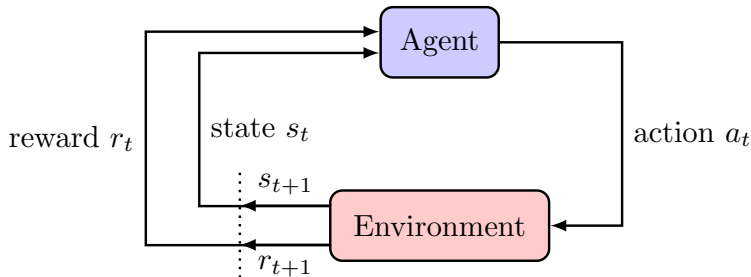
- RECAP
- LOCAL CONVERGENCE ANALYSIS
- NATURAL POLICY GRADIENT METHODS

# Outline

## ■ RECAP

# RL Framework

## ■ MARKOV DECISION PROCESSES (MDPs)



|                                                                                                                                                                                                                                    |                   |                                                                                                                                                                                                                          |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div style="border: 1px solid black; padding: 10px;"><p style="text-align: center; color: red; margin: 0;"><b>Goal</b></p><math display="block">\underset{\pi}{\text{maximize}} \quad \mathbb{E}_{s_0} [V^{\pi}(s_0)]</math></div> | $\Leftrightarrow$ | <div style="border: 1px solid black; padding: 10px;"><p style="text-align: center; color: red; margin: 0;"><b>Direct policy search</b></p><math display="block">\pi^+ \leftarrow \pi + \nabla_{\pi} V^{\pi}</math></div> |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

$\pi$  – a policy that maps states to actions

$V^{\pi}(s_0) := \mathbb{E} [\text{sum of all rewards received}]$

# Policy optimization

**Policy:**  $\pi : S \text{ (states)} \rightarrow \Delta_A \text{ (action distribution)}$

**Value function:**  $V^\pi(\rho) := \mathbb{E} \left[ \sum_{t=0}^{\infty} \gamma^t r(s_t, a_t) \mid s_0 \sim \rho \right]$

$P(\cdot \mid s, a) : S \times A \rightarrow \Delta_A$  – transition kernel

$s_0, a_0, r_0, s_1, a_1, \dots$  – a trajectory  $\tau$  generated by a policy  $\pi$

## ■ NONCONVEX POLICY OPTIMIZATION

$$\underset{\pi}{\text{maximize}} \quad V^\pi(\rho)$$

★ Search the decision in [a function space](#)

# Policy parametrization

**Policy:**  $\pi_{\theta}(\cdot | s): S \rightarrow \Delta_A$

**Parameter:**  $\theta \in \mathbb{R}^d$

## ■ TABULAR POLICY

$$\pi_{\theta}(a | s) = \theta_{s,a} \text{ w/ parameter } \theta \in \mathbb{R}^{|S||A|}$$

## ■ SOFTMAX POLICY

$$\pi_{\theta}(a | s) = \frac{e^{\theta_{s,a}}}{\sum_{a'} e^{\theta_{s,a'}}} \text{ w/ parameter } \theta \in \mathbb{R}^{|S||A|}$$

## ■ LOG-LINEAR POLICY

$$\pi_{\theta}(a | s) = \frac{e^{\theta^{\top} \phi_{s,a}}}{\sum_{a'} e^{\theta^{\top} \phi_{s,a'}}} \text{ w/ feature } \phi_{s,a} \in \mathbb{R}^d$$

# Policy gradient methods

**Policy:**  $\pi_{\theta}(\cdot | s): S \rightarrow \Delta_A$

**Parameter:**  $\theta \in \mathbb{R}^d$

## ■ NONCONVEX PARAMETER OPTIMIZATION

$$\underset{\theta}{\text{maximize}} \quad V^{\pi_{\theta}}(\rho)$$

## ■ GRADIENT ASCENT

$$\theta^+ \leftarrow \theta + \eta \nabla_{\theta} V^{\pi_{\theta}}(\rho)$$

★ Search the decision in a parameter space

## ■ EUCLIDEAN DISTANCE

$$\theta^+ \leftarrow \operatorname{argmax}_{\theta'} \langle \nabla_{\theta} V^{\pi_{\theta}}(\rho), \theta' - \theta \rangle - \frac{1}{2\eta} \|\theta' - \theta\|^2$$

- ★ Gradient ascent in Euclidean space

# Trajectory distribution and state visitation distribution

## ■ TRAJECTORY DISTRIBUTION

$$\Pr(\tau \mid \pi) = \rho(s_0)\pi(a_0 \mid s_0)P(s_1 \mid s_0, a_0)\pi(a_1 \mid s_1)P(s_2 \mid s_1, a_1) \cdots$$

$$\Pr(s_t = s \mid \pi) = \sum_{s_0, a_0, \dots, s_{t-1}, a_{t-1}} \Pr(s_0, a_0, \dots, s_{t-1}, a_{t-1}, s_t = s \mid \pi)$$

## ■ STATE VISITATION DISTRIBUTION

$$d_\rho^\pi(s) = (1 - \gamma) \sum_{t=0}^{\infty} \gamma^t \Pr(s_t = s \mid \pi)$$

## ■ LINEAR FORM OF VALUE FUNCTION

$$V^\pi(\rho) = \frac{1}{1 - \gamma} \mathbb{E}_{s \sim d_\rho^\pi(\cdot), a \sim \pi(\cdot \mid s)} [r(s, a)]$$

# Policy gradient

## ■ REINFORCE VERSION

$$\nabla_{\theta} V^{\pi_{\theta}}(\rho) = \mathbb{E}_{\tau} \left[ \left( \sum_{t=0}^{\infty} \gamma^t r_t \right) \left( \sum_{t=0}^{\infty} \nabla_{\theta} \log \pi_{\theta}(a_t | s_t) \right) \right]$$

## ■ STATE-ACTION VALUE VERSION

$$\nabla_{\theta} V^{\pi_{\theta}}(\rho) = \frac{1}{1-\gamma} \mathbb{E}_{s \sim d_{\rho}^{\pi_{\theta}}(\cdot), a \sim \pi_{\theta}(\cdot | s)} [Q^{\pi_{\theta}}(s, a) \nabla_{\theta} \log \pi_{\theta}(a | s)]$$

## ■ ADVANTAGE VERSION

$$\nabla_{\theta} V^{\pi_{\theta}}(\rho) = \frac{1}{1-\gamma} \mathbb{E}_{s \sim d_{\rho}^{\pi_{\theta}}(\cdot), a \sim \pi_{\theta}(\cdot | s)} [A^{\pi_{\theta}}(s, a) \nabla_{\theta} \log \pi_{\theta}(a | s)]$$

# Outline

- RECAP

- LOCAL CONVERGENCE ANALYSIS

# Stationary-point convergence

**Nonconvex function:**  $V^{\pi_\theta}(\rho)$

**Policy gradient:**  $\nabla_\theta V^{\pi_\theta}(\rho)$

**Gradient ascent:**  $\theta_{t+1} = \theta_t + \eta \nabla_\theta V^{\pi_{\theta_t}}(\rho)$

## ■ SMOOTH VALUE FUNCTION

$$\|\nabla_\theta V^{\pi_\theta}(\rho) - \nabla_\theta V^{\pi_{\theta'}}(\rho)\| \leq \beta \|\theta - \theta'\|$$

★ Lipschitz continuous gradient

# Quadratic bound

## ■ QUADRATIC UPPER BOUND

$$F(\theta) = \frac{\beta}{2}\|\theta\|^2 - V^{\pi_\theta}(\rho) \text{ is a convex function}$$

$\Downarrow$

$$V^{\pi_\theta}(\rho) - V^{\pi_{\theta'}}(\rho) - (\nabla_\theta V^{\pi_{\theta'}}(\rho))^\top (\theta - \theta') \leq \frac{\beta}{2}\|\theta - \theta'\|^2$$

## ■ QUADRATIC LOWER BOUND

$$\begin{aligned} V^{\pi_\theta}(\rho) - V^{\pi_{\theta'}}(\rho) - (\nabla_\theta V^{\pi_{\theta'}}(\rho))^\top (\theta - \theta') &\geq -\frac{\beta}{2}\|\theta - \theta'\|^2 \\ &\quad - (\nabla_\theta V^{\pi_{\theta'}}(\rho) - \nabla_\theta V^{\pi_\theta}(\rho))^\top (\theta - \theta') \\ &\geq -\frac{3\beta}{2}\|\theta - \theta'\|^2 \end{aligned}$$

## ■ QUADRATIC BOUND

$$\left| V^{\pi_{\theta}}(\rho) - V^{\pi_{\theta'}}(\rho) - (\nabla_{\theta} V^{\pi_{\theta'}}(\rho))^{\top} (\theta - \theta') \right| \leq \frac{3\beta}{2} \|\theta - \theta'\|^2$$

## ■ TELESCOPING SUM

$$\left( \nabla_{\theta} V^{(t)}(\rho) \right)^{\top} (\theta_{t+1} - \theta_t) \leq V^{(t+1)}(\rho) - V^{(t)}(\rho) + \frac{3\beta}{2} \|\theta_t - \theta_{t+1}\|^2$$

$\Downarrow$

$$\left( \eta - \frac{3\beta\eta^2}{2} \right) \frac{1}{T} \sum_{t=1}^T \left\| \nabla_{\theta} V^{(t)}(\rho) \right\|^2 \leq \frac{1}{T} \sum_{t=1}^T \left( V^{(t+1)}(\rho) - V^{(t)}(\rho) \right)$$

$\Downarrow (\eta = \frac{1}{2\beta})$

$$\frac{1}{T} \sum_{t=1}^T \left\| \nabla_{\theta} V^{(t)}(\rho) \right\|^2 \leq \frac{8\beta}{T} V^{(T+1)}(\rho)$$

★ Vanishing gradient of best-iterate

# Outline

- RECAP
- LOCAL CONVERGENCE ANALYSIS
- NATURAL POLICY GRADIENT METHODS

# Natural policy gradient

$$F_{\rho}^{\dagger}(\theta) \nabla_{\theta} V^{\pi_{\theta}}(\rho)$$

## ■ FISHER INFORMATION MATRIX

$$F_{\rho}(\theta) \quad := \quad \mathbb{E}_{s \sim d_{\rho}^{\pi_{\theta}}(\cdot)} \mathbb{E}_{a \sim \pi_{\theta}(\cdot | s)} \left[ \nabla_{\theta} \log \pi_{\theta}(a | s) (\nabla_{\theta} \log \pi_{\theta}(a | s))^{\top} \right]$$

$\nabla_{\theta} \log \pi_{\theta}(a | s)$  – Fisher score

$d_{\rho}^{\pi_{\theta}}(\cdot)$  – stationary state distribution

## ■ KL DIVERGENCE

$$\theta^+ \leftarrow \operatorname{argmax}_{\theta'} \langle \nabla_{\theta} V^{\pi_{\theta}}(\rho), \theta' - \theta \rangle - \frac{1}{\eta} D_{\text{KL}}(\pi_{\theta'} \parallel \pi_{\theta})$$

$$D_{\text{KL}}(\pi_{\theta'} \parallel \pi_{\theta}) = \mathbb{E}_{s \sim d_{\rho}^{\pi_{\theta}}(\cdot), a \sim \pi_{\theta'}(\cdot | s)} \left[ \log \frac{\pi_{\theta'}(a | s)}{\pi_{\theta}(a | s)} \right] \quad - \text{ weighted KL divergence}$$

### ★ Quadratic approximation

$$\begin{aligned} D_{\text{KL}}(\pi_{\theta'} \parallel \pi_{\theta}) &\approx (\nabla_{\theta'} D_{\text{KL}}(\pi_{\theta'} \parallel \pi_{\theta})|_{\theta'=\theta})^{\top} (\theta' - \theta) \\ &\quad + \frac{1}{2} (\theta' - \theta)^{\top} \nabla_{\theta'}^2 D_{\text{KL}}(\pi_{\theta'} \parallel \pi_{\theta})|_{\theta'=\theta} (\theta' - \theta) \\ &= \frac{1}{2} (\theta' - \theta)^{\top} F_{\rho}(\theta) (\theta' - \theta) \end{aligned}$$

### ★ Gradient ascent in distribution space

# Natural policy gradient methods

**Policy:**  $\pi_{\theta}(\cdot | s): S \rightarrow \Delta_A$

**Parameter:**  $\theta \in \mathbb{R}^d$

## ■ NONCONVEX PARAMETER OPTIMIZATION

$$\underset{\theta}{\text{maximize}} \quad V^{\pi_{\theta}}(\rho)$$

## ■ NATURAL GRADIENT ASCENT

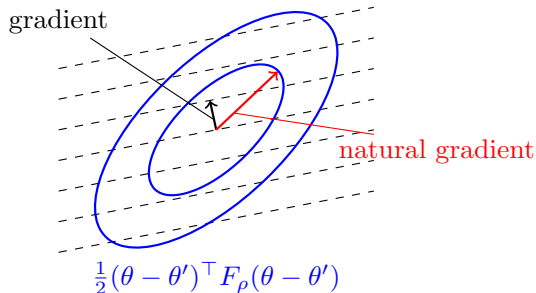
$$\theta^+ \leftarrow \theta + \eta F_{\rho}^{\dagger}(\theta) \nabla_{\theta} V^{\pi_{\theta}}(\rho)$$

- ★ Search the decision in **a parameter space**
- ★ **Costly inverse** of Fisher information matrix

# Natural policy gradient

**Natural policy gradient:**  $F_{\rho}(\theta)^{\dagger} \nabla_{\theta} V^{\pi_{\theta}}(\rho)$

**Fisher information:**  $F_{\rho}(\theta) = \mathbb{E} \left[ \nabla_{\theta} \log \pi_{\theta} (\nabla_{\theta} \log \pi_{\theta})^{\top} \right]$



**Steepest ascent** in Fisher information distance

# Natural policy gradient: multiplicative weights update

$$\pi_{\theta}^{+}(a | s) = \pi_{\theta}(a | s) \frac{\exp\left(\frac{\eta}{1-\gamma} A^{\pi_{\theta}}(s, a)\right)}{Z^{\pi_{\theta}}(s)}$$

$Z^{\pi_{\theta}}(s) := \sum_{a \in A} \pi_{\theta}(a | s) \exp\left(\frac{\eta}{1-\gamma} A^{\pi_{\theta}}(s, a)\right)$  – normalization constant

$A^{\pi_{\theta}}(s, a) = Q^{\pi_{\theta}}(s, a) - V^{\pi_{\theta}}(s)$  – advantage value function

## ■ NATURAL POLICY GRADIENT

$$\begin{aligned}(1 - \gamma) F_{\rho}^{\dagger}(\theta) \nabla_{\theta} V^{\pi_{\theta}}(\rho) &= F_{\rho}^{\dagger}(\theta) \mathbb{E}_{s \sim d_{\rho}^{\pi_{\theta}}(\cdot), a \sim \pi_{\theta}(\cdot | s)} [A^{\pi_{\theta}}(s, a) \nabla_{\theta} \log \pi_{\theta}(a | s)] \\ &= \operatorname{argmin}_w \mathbb{E}_{s \sim d_{\rho}^{\pi_{\theta}}(\cdot), a \sim \pi_{\theta}(\cdot | s)} \left[ \left( A^{\pi_{\theta}}(s, a) - w^{\top} \nabla_{\theta} \log \pi_{\theta}(a | s) \right)^2 \right]\end{aligned}$$

# Summary

## ■ RECAP

Policy gradient methods

## ■ LOCAL CONVERGENCE ANALYSIS

Local best-iterate convergence

## ■ NATURAL POLICY GRADIENT METHODS

Multiplicative weights update